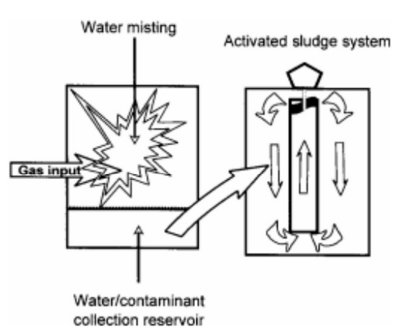
CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
III Semester of MSc Biotechnology Examination- November, 2019
Course code & Name: BT823: Environmental Biotechnology
Date: 15.11.2019 Day: Friday Time: 01.30 p.m to 02.00 pm Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	20
1.	Ozone layer is found in: (i) trophposphere (ii) stratosphere (iii) thermosphere (iv) ionosphere	
2.	The below given biochemical reaction is part of which of the process occurring in ecosystems? $NH_4^+ + 1.5O_2 \rightarrow NO_2^-$ (i) nitrification (ii) ammonification (iii) denitrification (iv) all of the above	
3.	The following figure correctly depicts which of the biotechnology based pollution control technology?  (i) Biotrickling filter (ii) Bioscrubber (iii) Biofilter (iv) none of the above	

4.	The statement “If humans identify with nature, then taking care of the natural world will become part of taking care of one’s self” denotes (i) Ecofeminism (ii) Deep ecology (iii) eco-centric holism (iv) patriarchal dualism	
5.	Biotransformation rate of hexacholorcyclohexane in field, land farming, slurry bioreactor and laboratory reactor is in the following order: (i) field> land farming>slurry bioreactor> laboratory reactor (ii) field< land farming<slurry bioreactor< laboratory reactor (iii) field< land farming>slurry bioreactor< laboratory reactor (iv) field= land farming>slurry bioreactor> laboratory reactor.	
6.	During biodegradation of toluene, the products of complete mineralization of toluene in denitrifying zones would include: (i) CO_2, H_2O, N_2 (ii) CO_2, H_2O, NO_2 (iii) CO_2, H_2O, NH_3 (iv) CO_2, H_2O, NH_4OH	
7.	Bioremediation can clean up polluted soils by: (i) adding nutrients to stimulate the activity of certain soil bacteria (ii) inoculating the soil with certain bacteria that can degrade toxic organic compounds (iii) using plants to stimulate soil microbial activity by exuding energy compounds into the rhizosphere (iv) all of the above	
8.	The feasibility of *in situ* bioremediation depends on (i) hydraulic conductivity (ii) sorption capacity of the aquifer materials (iii) potential biodegradation rate (iv) All of the above	
9.	Rhamnolipids belong to which class of biosurfactants? (i) glycolipids (ii) phospholipids (iii) polymeric biosurfactants (iv) lipopeptides	
10.	Dual phase extraction system is also known as (i) Bioslurping (ii) Bioventing (iii) Biobarriers (iv) biofilers	
11.	Bioventing is a suitable *in situ* bioremediation strategy for which of the following class of pollutants? (i) heavy metals (ii) BTEX compounds (iii) Azo dyes (iv) Perchloroethylene	
12.	Lignin is: (i) xenobiotic and persistent compound (ii) xenobiotic compound (iii) both xenobiotic and recalcitrant compound (iv) both recalcitrant and persistent compound	

13.	Hexane biodegradation involves which of the following enzyme(s)? (i) alkane monooxygenase (ii) alcohol dehydrogenase (iii) aldehyde dehydrogenase (iv) all of the above	
14.	Acetaldehyde and pyruvic acid are the end products of which of the following pathways? (i) o- cleavage (ii) m-cleavage of catechol (iii) homogentisate pathway (iv) none of the above	
15.	Biologically active zones that are placed in the path of (preferably shallow and narrow) BTEX plumes are known as (i) Biofilters (ii) bioscrubbers (iii) Biobarriers (iv) Biotrickling filters	
16.	MEOR is a method used for (i) oil recovery (ii) biodiesel production (iii) Metal leaching (iv) bioremediation	
17.	Which of the following reactor uses anaerobic sludge? (i) Trickling filter (ii) UASB (iii) anaerobic pond (iv) Imhoff tank	
18.	With reference to activated sludge process for treatment of wastewater, in the expression, HRT=24 .V_{at}/Q, Q denotes: (i) flow rate of wastewater (ii) volume of wastewater (iii) volume of aeration tank (iv) temperature of wastewater	
19.	White rot fungi are useful for bioremediation of contaminated soil due to production of which of the following enzymes? (i) Lignin peroxidases (LiP) (ii) Manganese peroxidases (MnP) (iii) Laccases (iv) All of the above	
20.	The product of following reaction involved in bioleaching of minerals is: $2FeS_2 + 7O_2 + 2H_2O \xrightarrow{\text{thiobacilli}} + 2H_2SO_4$ (i) H_2S (ii) 2 FeSO4 (iii) H_2SO_4 (iv) all of the above	

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Date: 15.11.2019 Day: Friday Time: 02.00 p.m to 04.30 pm Maximum Marks: 50

Instructions:
1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

	SECTION -I	
Q - II Answer the following questions as directed		**20**
Q-II (A)	Answer **any one** of the following:	**05**
	(i) Discuss the use of any two biotechnological methods for treatment of polluted air. (ii) Giving major biochemical reactions and microorganisms involved explain importance of nitrification and denitrification reactions.	
Q-II (B)	Answer **any one** of the following	**05**
	(i) Discuss the role of any two classes of enzymes in biodegradation of xenobiotics. (ii) Discuss how bioavailability is an important parameter influencing biodegradation of xenobiotics in soil.	
Q-II (C)	Answer **any one** of the following:	**05**
	(i) Critically comment: Economic development and environmental protection should have synergistic relationship. (ii) Describe the ortho and meta cleavage pathways for biodegradation of aromatic hydrocarbons.	
Q-II (D)	Answer **any one** of the following	**05**
	(i) Briefly discuss phytovolatilization and phytoextraction techniques for bioremediation of contaminated soils. (ii) Discuss how the characteristics of contaminated site influences success of bioremediation strategies?	

SECTION - II		
Q-III Answer the following questions as directed		**30**
QIII-(A)	Answer **any one** of the following	**05**
	(i) Discuss the importance of Monod equation and dual resistance model in biodegradation of pollutants. (ii) Discuss the role of various products of microorganisms in microbially enhanced oil recovery (MEOR)?	
Q-III (B)	Discuss **any one** of the following:	**05**
	(i) Any two industrial scale bioleaching techniques (ii) Strategies for heavy metal bioremediation	
Q-III (C)	Giving diagrams discuss the use of **any two** of the following:	**10**
	(i) Rotating biological contactors (ii) Upflow anaerobic sludge blanket (UASB) reactor (iii) Activated sludge process	
Q-III (D)	Anwer **any two** of the following	**10**
	(i) What are biofuels? Write a note on biofluels. (ii) Describe the use of xylanases in paper pulp processing. (iii) Discuss the mechanisms of bioleaching.	